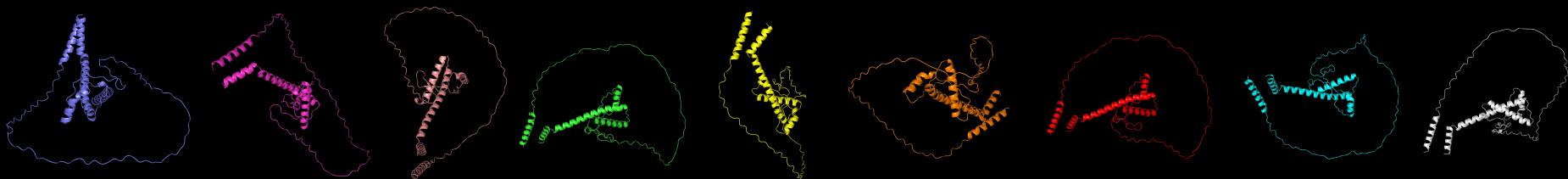


Causal Inference with Protein Sequences: Biologically Plausible Dimension Reduction Using AlphaFold

Matthew D. Laws, Williams College



Collaborators

Zhonghua Liu

Biostatistics, Columbia University



Rohit Bhattacharya

Computer Science, Williams College



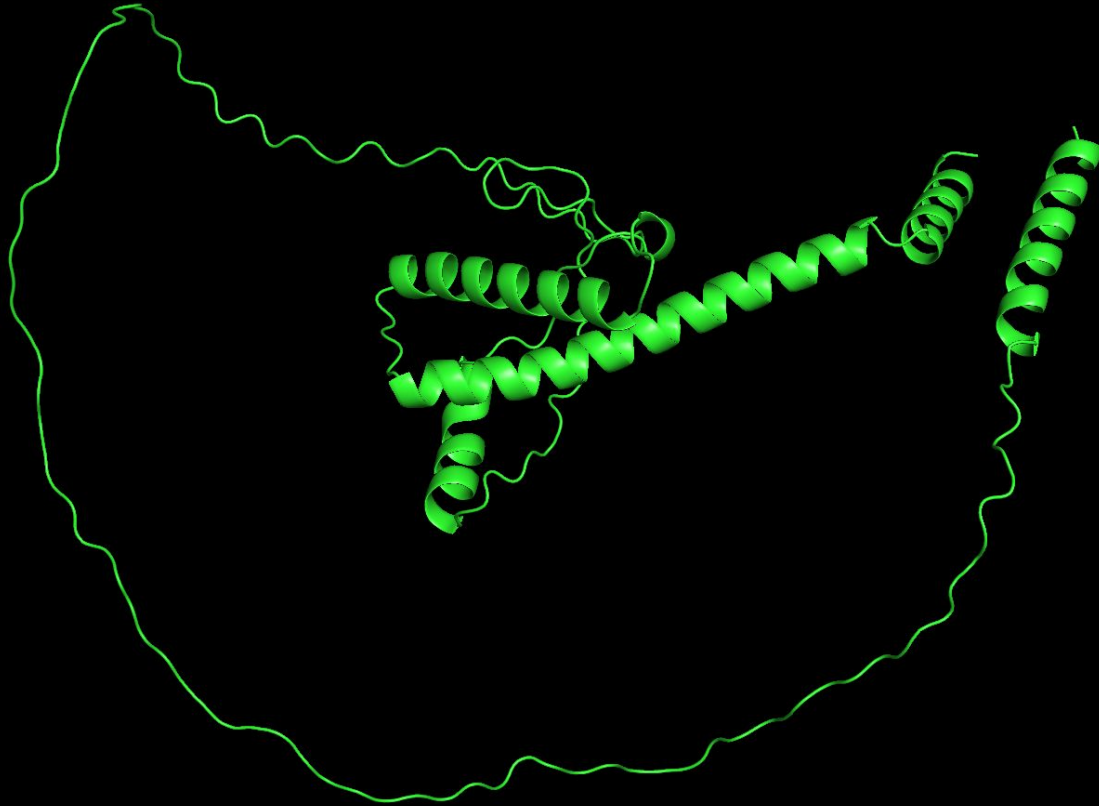
Motivating Question

Can we determine causal relationships between
diseases and **mutated protein sequences**
through observational causal inference?

Motivating Example: Prion Protein

10	20	30	40	50	60	70
MANLGCWMLV	LFVATWSDLG	LCKKRPKPGG	WNTGGSRYPG	QGSPGGNRYP	PQGGGGWGQP	HGGGWGQPHG
80	90	100	110	120	130	140
GGWGQPHGGG	WGQPHGGGWG	QGGGTHSQWN	KPSKPKTNMK	HMAGAAAAGA	VVGGLGGYML	GSAMSRPIIH
150	160	170	180	190	200	210
FGSDYEDRY	RENMHRYPNQ	VYYRPMDEYS	NQNNFVHDCV	NITIKQHTVT	TTTKGENFTE	TDVKMMERVV
220	230	240	250	253		
EQMCITQYER	ESQAYYQRGS	SMVLFSSPPV	ILLISFLIFL	IVG		

Motivating Example: Prion Protein





ATOM 1	N N	.	MET A 1 1	?	31.591	17.450	-55.045	1.00	51.31	1	A 1
ATOM 2	C CA	.	MET A 1 1	?	30.667	16.954	-54.005	1.00	62.53	1	A 1
ATOM 3	C C	.	MET A 1 1	?	31.375	16.354	-52.782	1.00	63.64	1	A 1
ATOM 4	O O	.	MET A 1 1	?	30.803	16.379	-51.709	1.00	59.30	1	A 1
ATOM 5	C CB	.	MET A 1 1	?	29.635	15.995	-54.605	1.00	57.80	1	A 1
ATOM 6	C CG	.	MET A 1 1	?	28.739	16.727	-55.630	1.00	52.83	1	A 1
ATOM 7	S SD	.	MET A 1 1	?	27.351	15.716	-56.181	1.00	49.58	1	A 1
ATOM 8	C CE	.	MET A 1 1	?	26.692	16.715	-57.532	1.00	41.44	1	A 1
ATOM 9	N N	.	ALA A 1 2	?	32.637	15.902	-52.895	1.00	61.29	2	A 1
ATOM 10	C CA	.	ALA A 1 2	?	33.396	15.310	-51.777	1.00	65.09	2	A 1
ATOM 11	C C	.	ALA A 1 2	?	33.627	16.256	-50.572	1.00	65.34	2	A 1
ATOM 12	O O	.	ALA A 1 2	?	33.622	15.813	-49.426	1.00	59.17	2	A 1
ATOM 13	C CB	.	ALA A 1 2	?	34.728	14.797	-52.339	1.00	60.35	2	A 1
ATOM 14	N N	.	ASN A 1 3	?	33.766	17.556	-50.808	1.00	54.34	3	A 1
ATOM 15	C CA	.	ASN A 1 3	?	34.022	18.512	-49.721	1.00	60.79	3	A 1
ATOM 16	C C	.	ASN A 1 3	?	32.822	18.721	-48.783	1.00	61.08	3	A 1
ATOM 17	O O	.	ASN A 1 3	?	33.020	18.901	-47.583	1.00	59.12	3	A 1
ATOM 18	C CB	.	ASN A 1 3	?	34.527	19.836	-50.331	1.00	56.32	3	A 1
ATOM 19	C CG	.	ASN A 1 3	?	35.976	19.745	-50.783	1.00	51.14	3	A 1
ATOM 20	O OD1	.	ASN A 1 3	?	36.785	19.052	-50.193	1.00	49.01	3	A 1
ATOM 21	N ND2	.	ASN A 1 3	?	36.358	20.412	-51.846	1.00	48.08	3	A 1

Traditional Assumptions for Identification

Consistency:

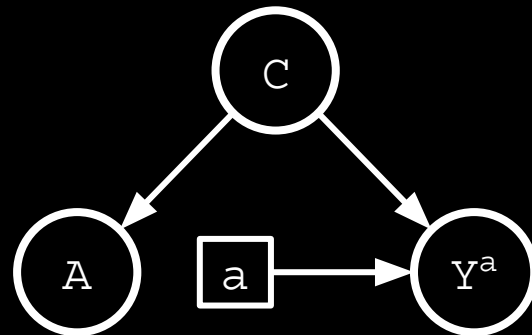
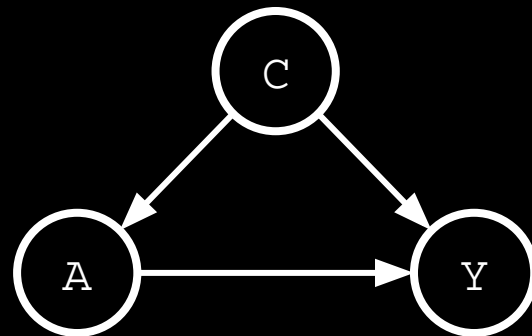
$$Y = Y^a \text{ when } A = a$$

Positivity:

$$P(A = a \mid C = c) > 0 \text{ for all } a, c$$

Conditional Ignorability:

$$Y^a \perp\!\!\!\perp A \mid C$$



Traditional Assumptions for Identification

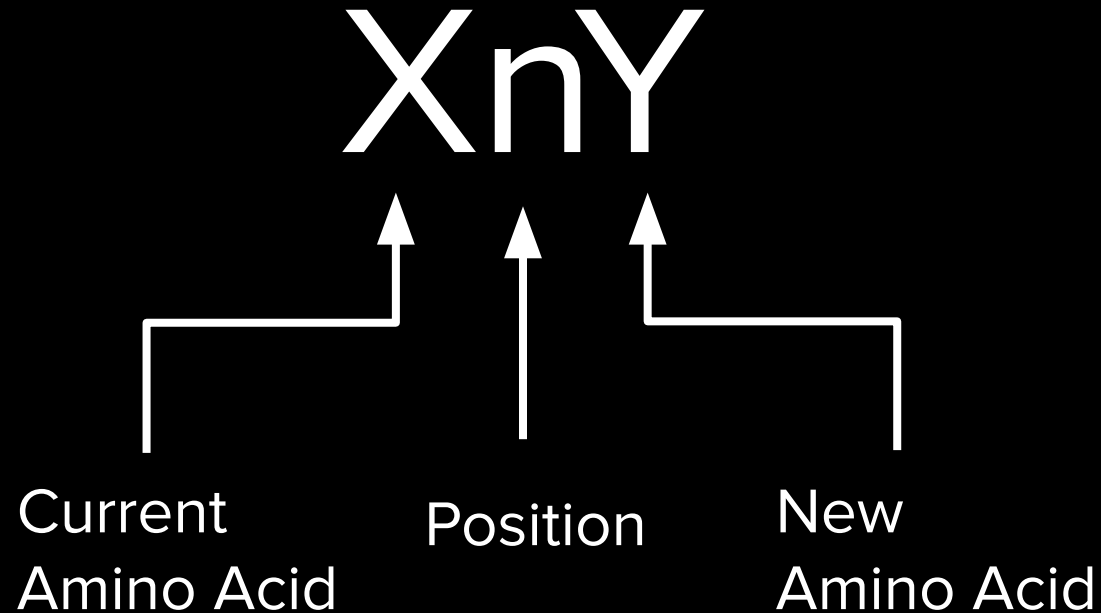
If we assume A to be binary then identification looks like:

$$\text{ACE} = \mathbb{E} [\mathbb{E}[Y \mid A = 1, C] - \mathbb{E}[Y \mid A = 0, C]]$$

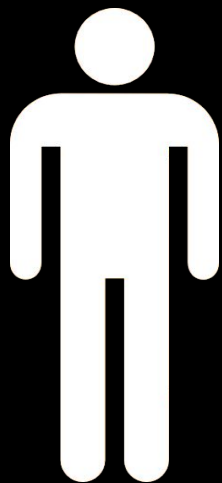
Roadmap: Different Treatment Encodings

Encoding	Consistency	Positivity	Biological Complexity
Binary treatment	✗	✓	✗
Raw amino acid sequence	✓	✗	✓
3D protein structure	✓	✗	✓ ⁺
Continuous misalignment score	✓	✓	✓

Mutation Notation



Binary Treatment: Consistency Violation

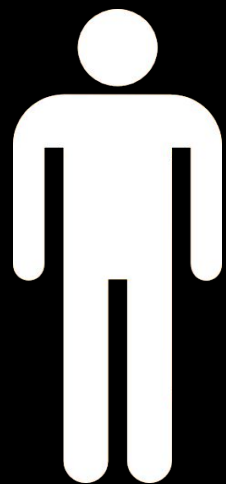


M129V

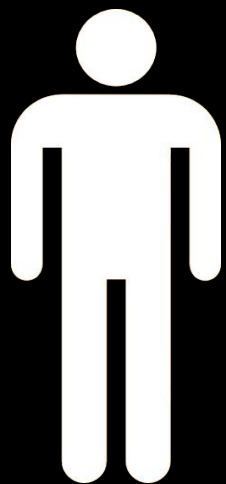


M129V + D178N

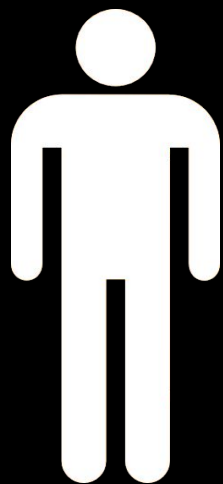
Unique Treatment per Sequence: Positivity Violation



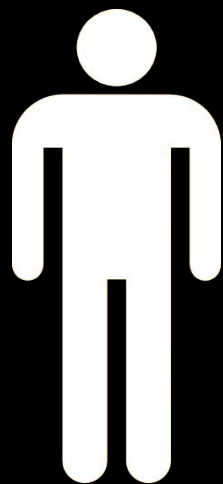
M129V



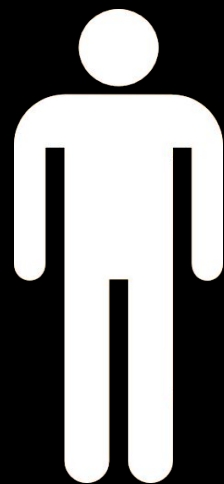
M129V
D178N



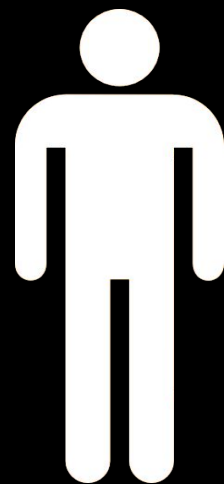
G127V



M232K



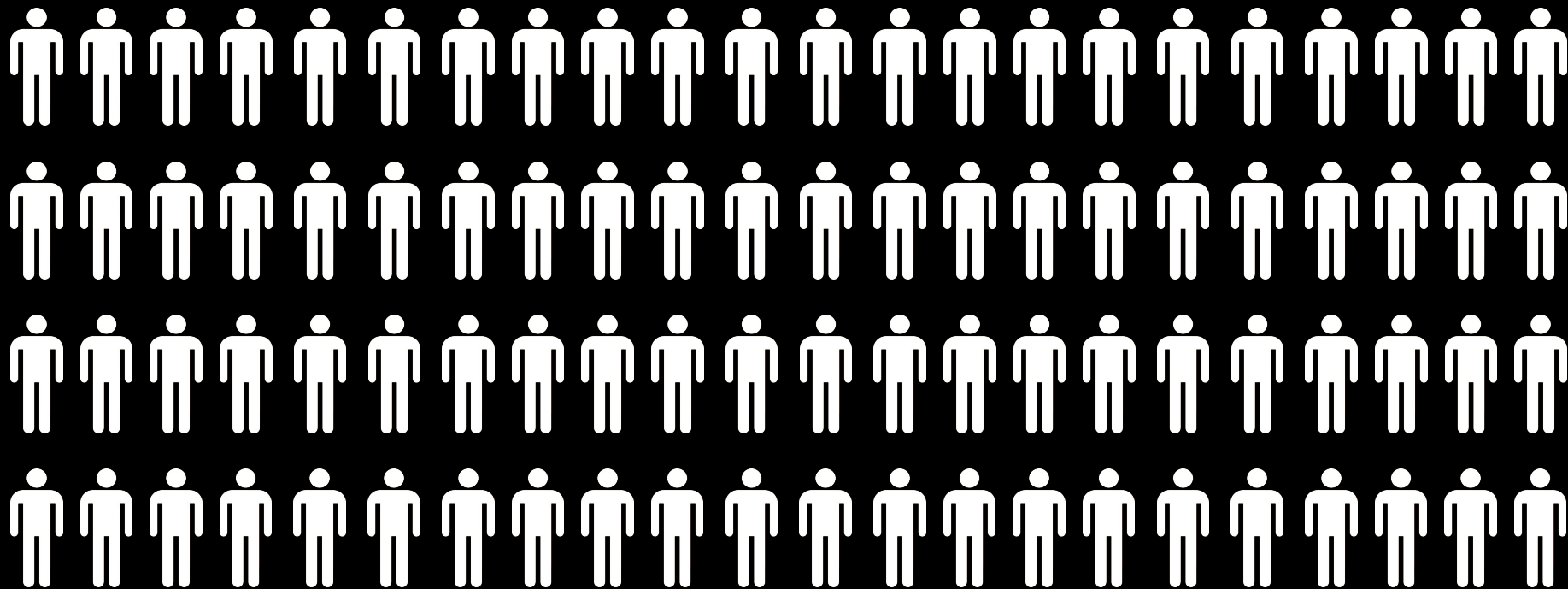
P102L



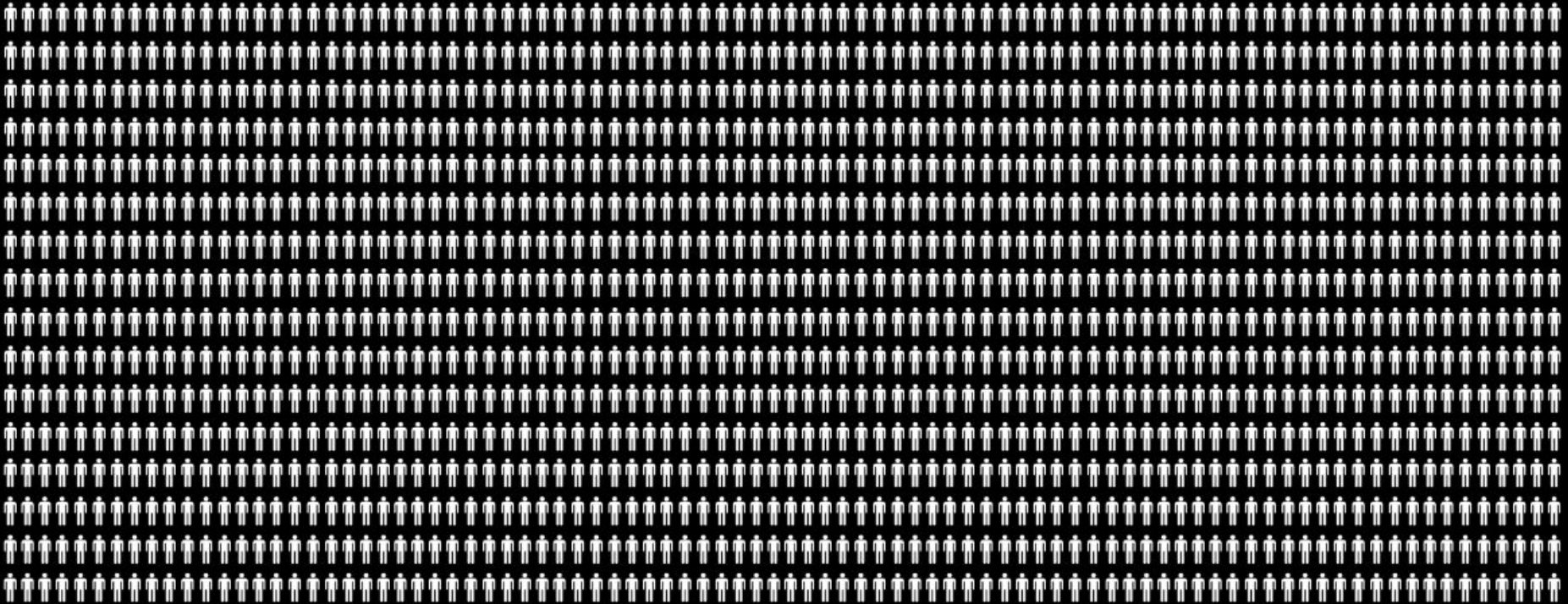
None

...

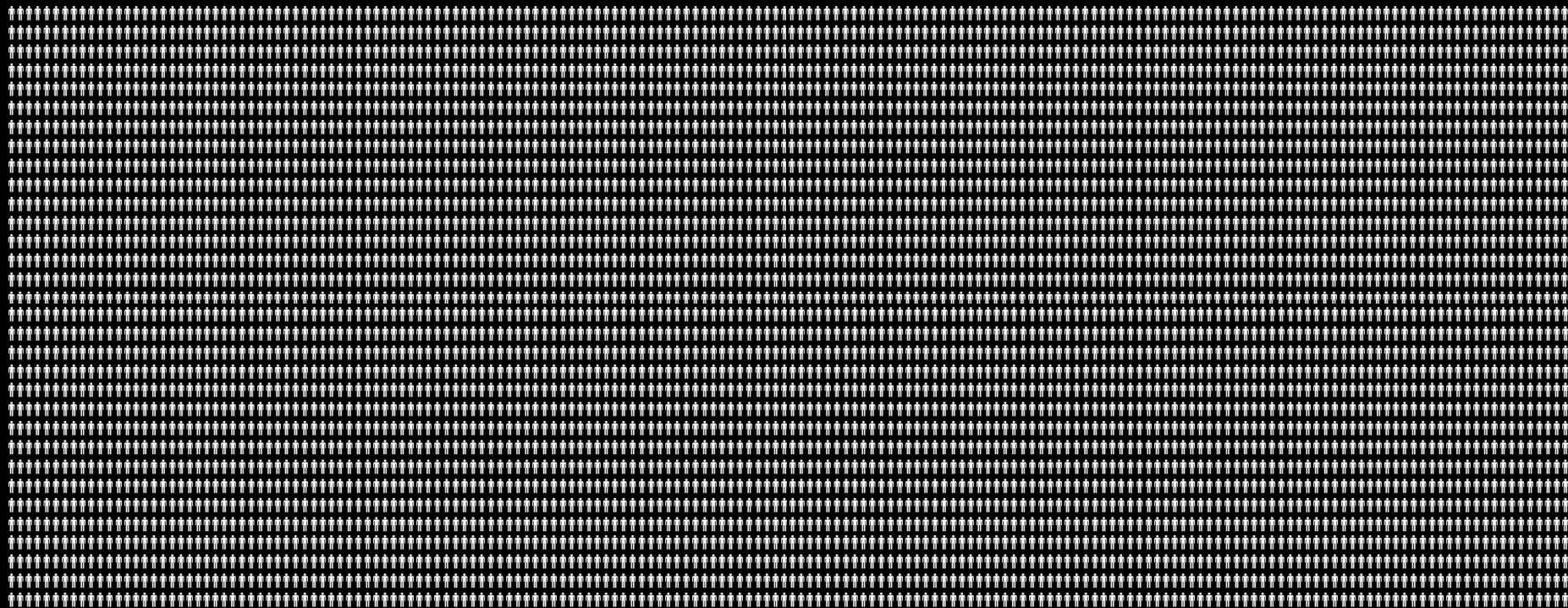
Unique Treatment per Sequence: Positivity Violation



Unique Treatment per Sequence: Positivity Violation



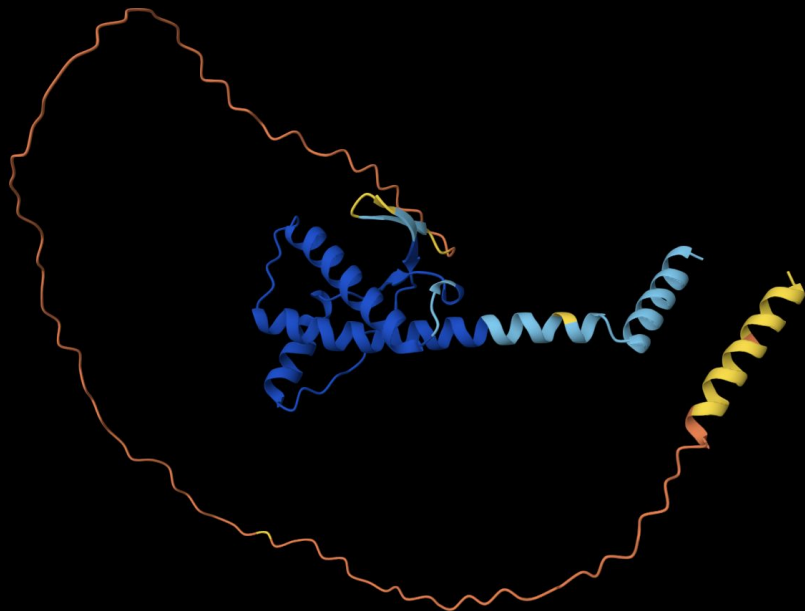
Unique Treatment per Sequence: Positivity Violation



Where do we go from here?

AlphaFold

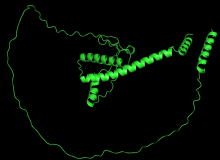
- Deep Learning Model
- Takes a **protein sequence** as input
- Outputs **3D coordinates** of all atoms in protein
- Won the 2024 Nobel Prize for Chemistry



Mutation(s) Structure (F) Disease



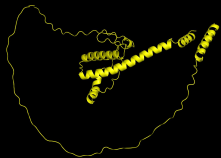
None



No



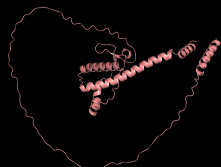
D178N



No



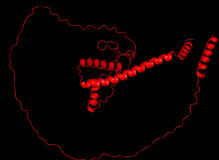
D178N
M129V



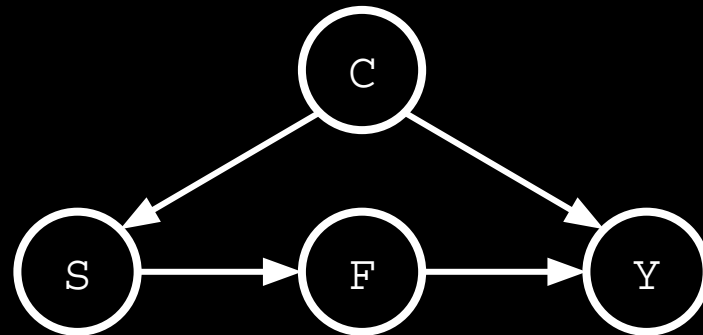
Yes



M129V



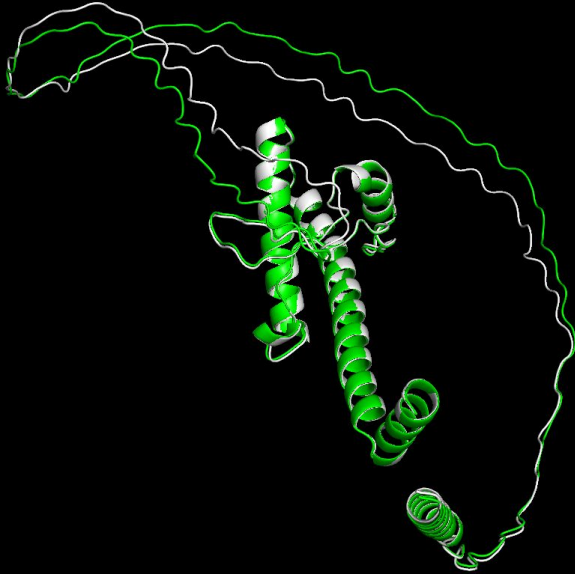
Yes



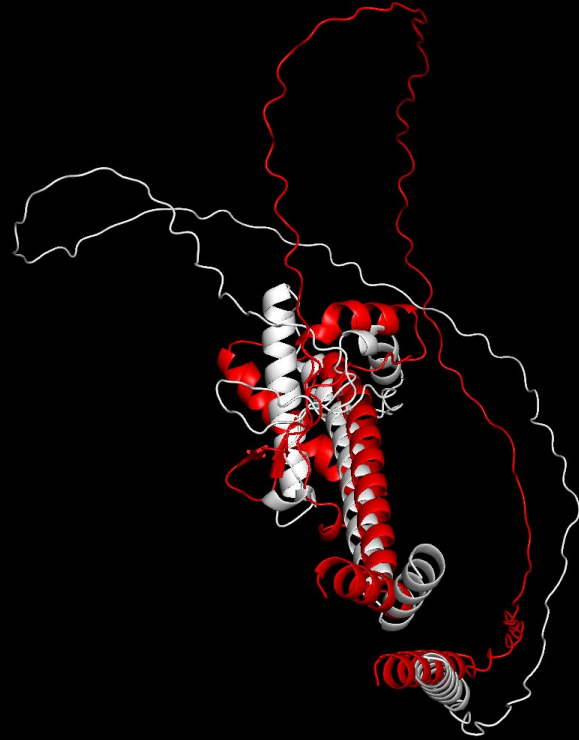
Positivity
Violation!

How do we encode 3D
structure numerically?

Protein Alignment

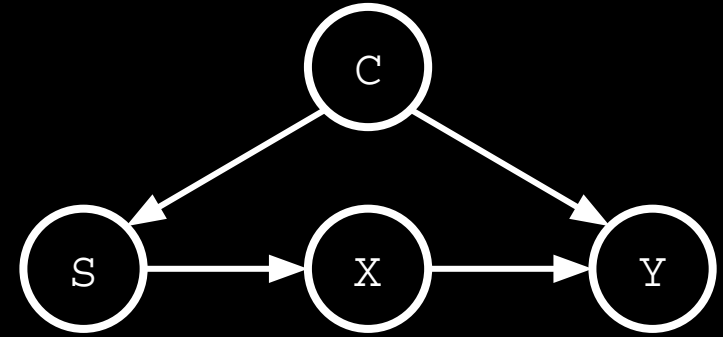


Benign Alignment



Pathogenic Alignment

	Mutation(s)	Misalignment (x)	Disease
	None	0.00	No
	D178N	0.89	No
	D178N M129V	2.57	Yes
	M129V	2.56	Yes



Consistency Violation?
No!

Positivity Violation?
No! (We can leverage
existing techniques)

Causal Isotonic Regression¹

$$\Gamma(a) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{(-\infty, a]}(A_i) \left[\frac{y_i - \mu(a_i, c_i)}{g(a_i, c_i)} \right] + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{I}_{(-\infty, a]}(A_i) \mu(a_i, c_j)$$

Then we find the Greatest Convex Minorant (GCM) for points $(\mathbf{F}(a), \Gamma(a))$ for all values of a where $\mathbf{F}$ is the empirical distribution

We assume that probability of disease is monotonically increasing with respect to misalignment

1. Westling et al 2020 Journal of the Royal Statistical Society: Series B (Statistical Methodology)

Inference

Pre-intervention

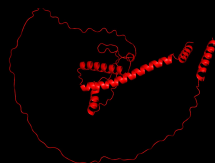
With M129V intervention

Pre-intervention		With M129V intervention		
Mutation(s)	Mutation(s)	Structure	Misalign	P(Disease ^{M129V})



None

M129V



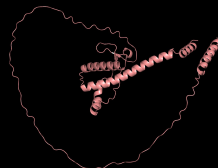
2.56

0.21



D178N

D178N
M129V



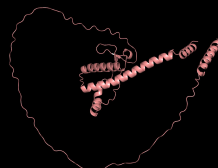
2.57

0.22



D178N
M129V

D178N
M129V



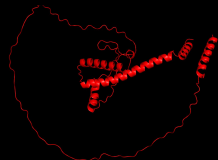
2.57

0.22



M129V

M129V



2.56

0.21

Pre-intervention

Without M129V intervention

Mutation(s)

Mutation(s)

Structure

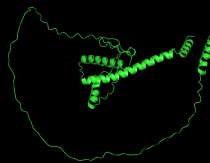
Misalign

P(Disease^{no M129V})



None

None



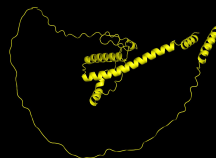
0.00

0.00



D178N

D178N



0.89

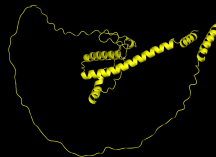
0.10



D178N

M129V

D178N



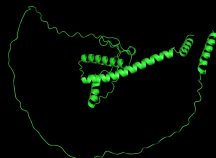
0.89

0.10



M129V

None



0.00

0.00

with M129V

without M129V

	Mutation(s)	Misalign	P(Disease ^{M129V})	Mutation(s)	Misalign	P(Disease ^{no M129V})
--	-------------	----------	------------------------------	-------------	----------	---------------------------------



M129V

2.56

0.21

None

0.00

0.00



D178N

2.57

0.22

D178N

0.89

0.10

M129V



D178N

2.57

0.22

D178N

0.89

0.10

M129V



M129V

2.56

0.21

None

0.00

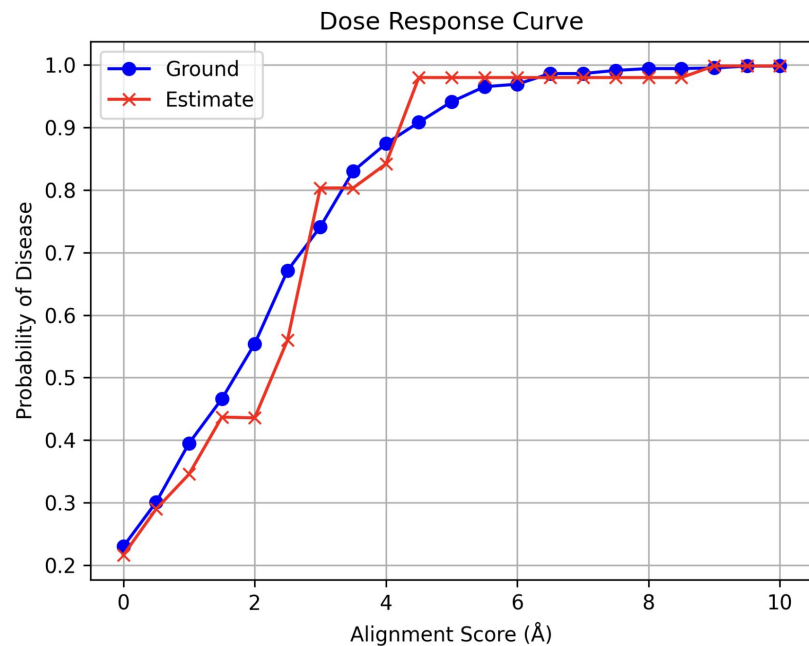
0.00

Results

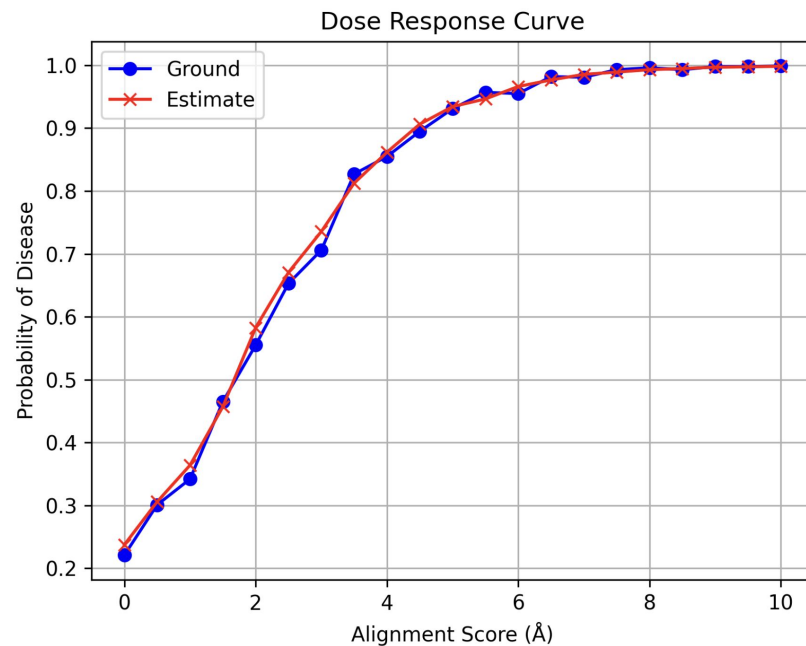
Data

Total Mutations	34 (100%)
Pathogenic Mutations	25 (73.5%)
Benign Mutations	9 (26.5%)

Dose Response Curve

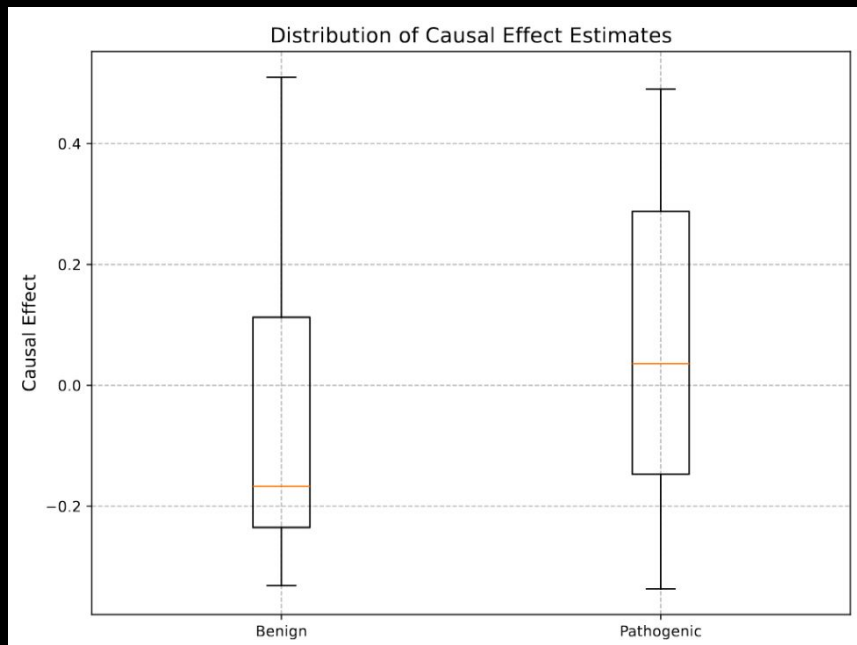


Dose Response Curve for Uniprot mutations

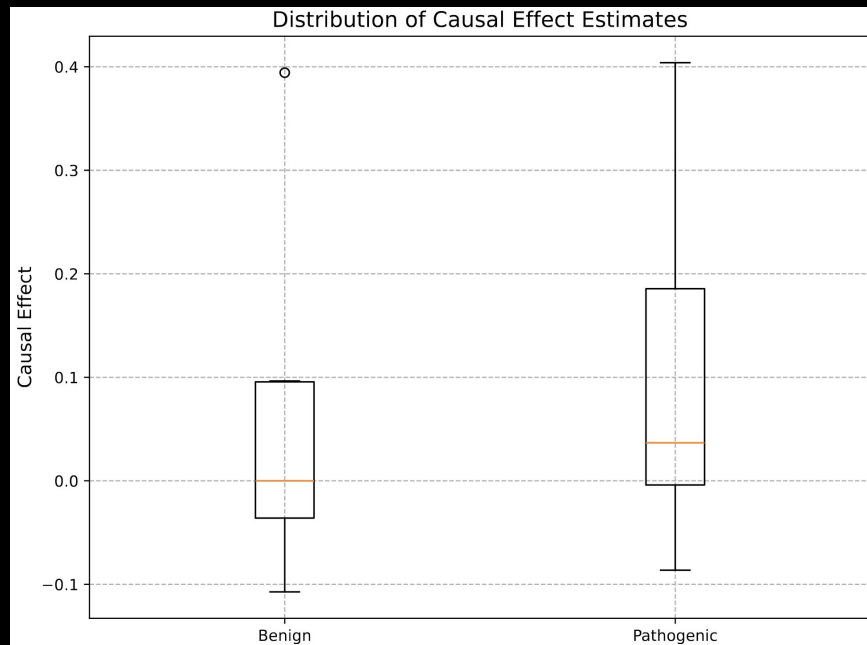


Dose Response Curve adding unstudied mutations

Distribution of Benign and Pathogenic Mutations



Baseline



Our Method

Confusion Matrix @ 0.0 Threshold

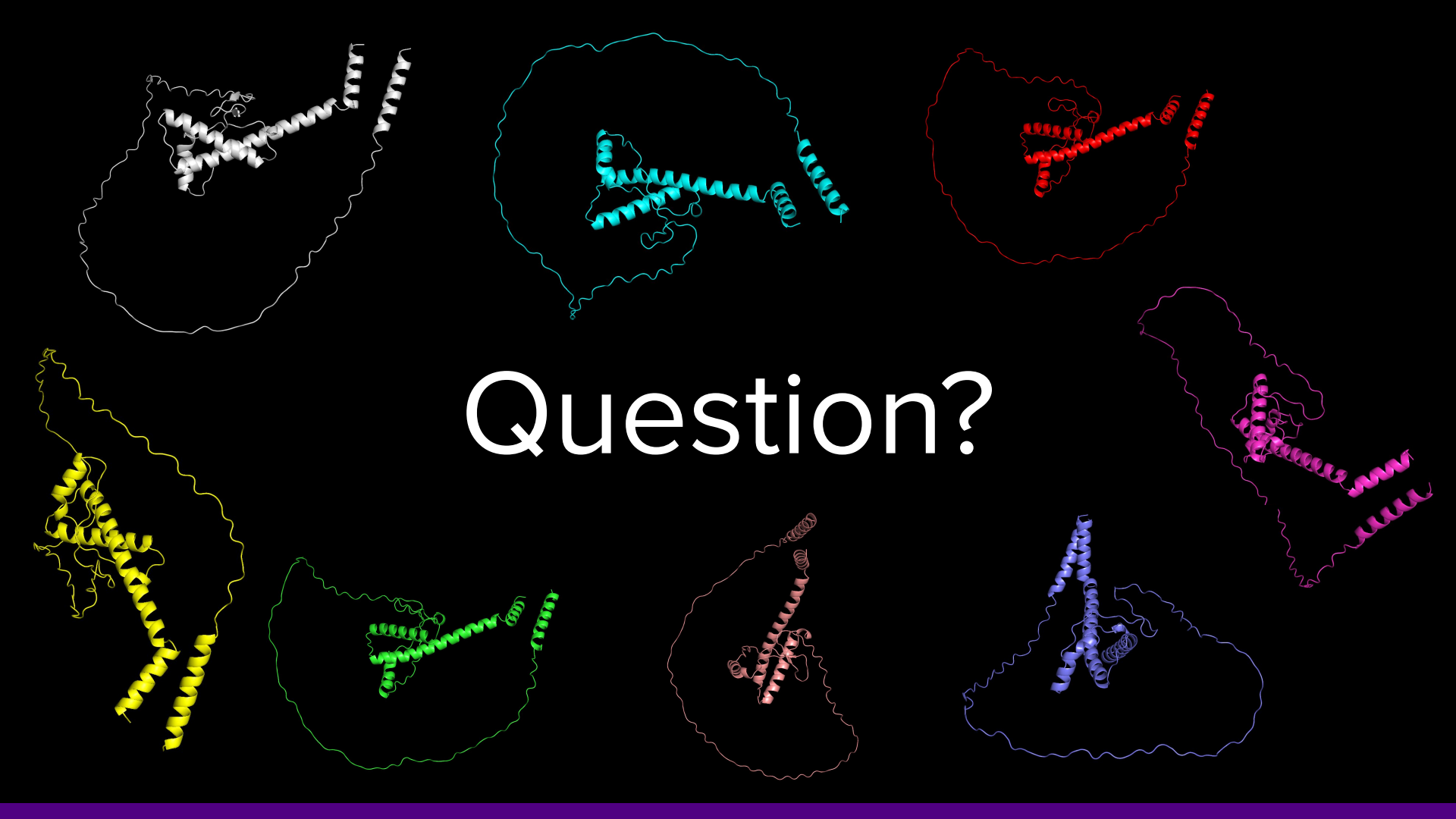
True Positives: 17	False Positives: 5
False Negatives: 8	True Negatives: 4

Confusion Matrix @ 0.1 Threshold

True Positives: 10	False Positives: 1
False Negatives: 15	True Negatives: 8

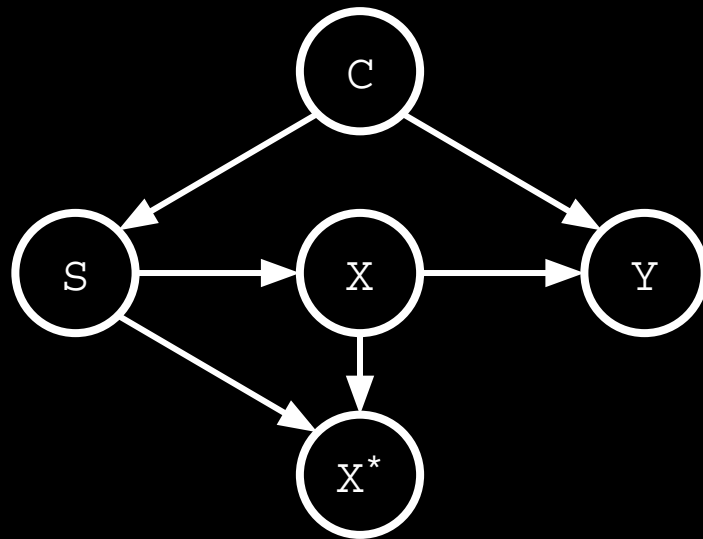
Future Work

- Working on gaining access to the UK BioBank to run real world experiments.
- Explore how other factors affect function besides structure, and how can we form a treatment variable for them.
- Continue to loosen the assumptions made by our proposed setup.
- Validate results in wet lab using CRISPR knockout.

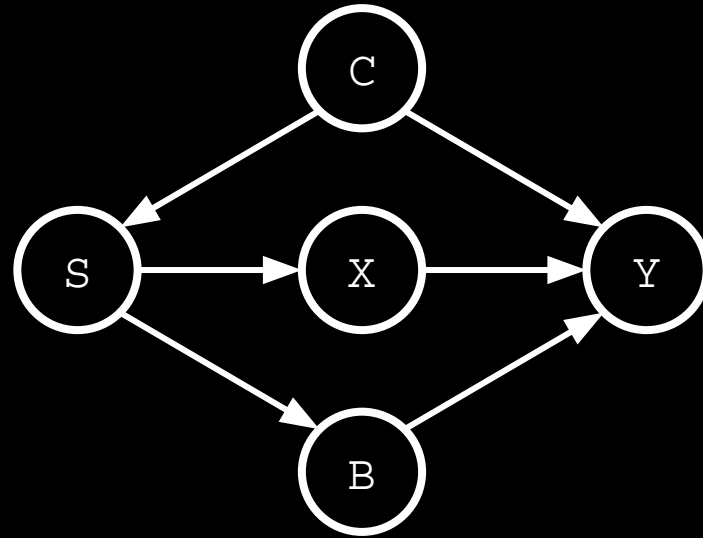


Question?

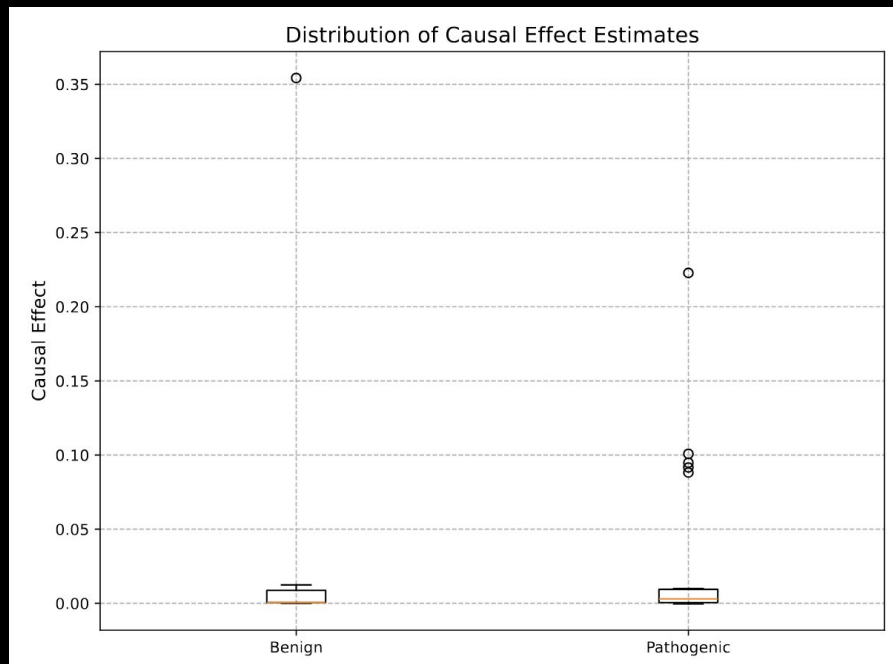
Measurement Error



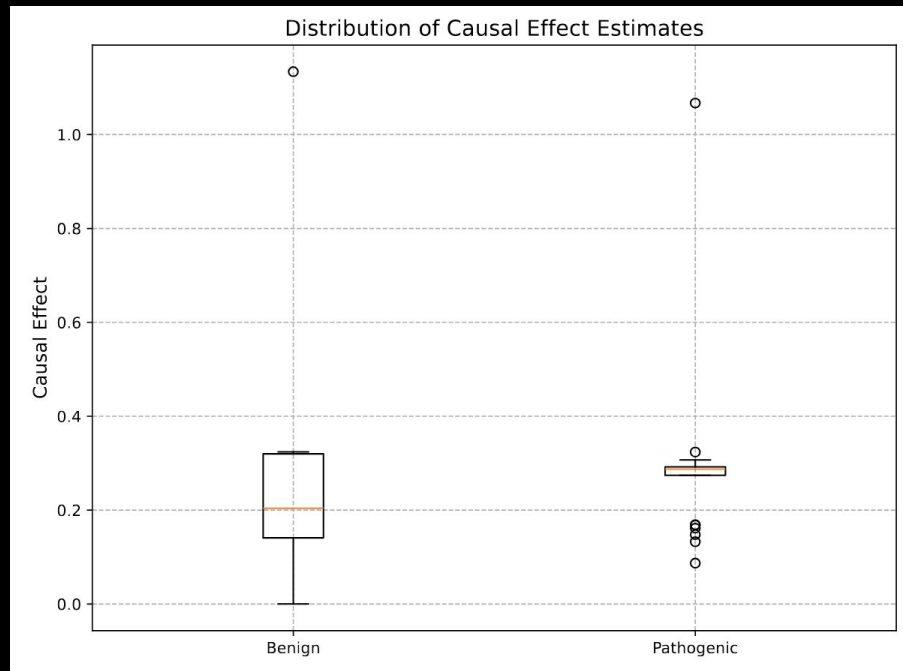
Multiple Effects



Sequence Only Adjustment



ProteinBERT Scoring



BLOSUM Scoring